

KB-001-Session-Start-End

KB-001 — Session Won't Start / Session Ended Early

Category: Session Issues

Call Volume: ~27% of all support contacts

Resolution Rate on Call: ~95%

Primary Remote Action: Soft reset, Hard reset

Why This Matters

Session start and premature-end failures are the single highest-volume support category. The good news: approximately 27% of these resolve with a remote reset alone, making them highly automatable. Understanding the failure chain — from driver tap to active charging — lets agents triage quickly and avoid unnecessary technician dispatches.

Background: How a Charging Session Actually Starts

Understanding the handshake helps agents reason about where in the chain a failure occurred.

1. **Driver presents payment or credential** (app, RFID card, tap-to-pay, plug-and-charge)
2. **Network authorizes the session** — the charger contacts the OCPP backend to validate payment and unlock the connector
3. **Connector lock engages** — the physical latch locks the cable into the vehicle inlet
4. **ISO 15118 / OCPP handshake** — the charger and vehicle negotiate power levels, charge limits, and session parameters
5. **Power flow begins** — DC or AC current flows; the session is live

A failure at any step produces a different symptom. If the agent knows where it broke, they skip unnecessary steps.

Section A — Session Won’t Start

Common Root Causes

Cause	Frequency	Identifiable by
Loose or unseated cable connection	~15%	Resolves on re-plug
OCPD authorization timeout or backend error	~20%	Remote reset resolves
Payment/credential authorization failure	~18%	Error code on screen; see KB-002
Vehicle-side communication fault	~12%	Multiple chargers fail for same vehicle
Charger hardware fault (relay, cable, port)	~10%	Persists after reset; see KB-003
Ghost-online / stale telemetry	~8%	App shows available, charger unresponsive; see KB-004
Software/firmware crash	~7%	Hard reset resolves
RFID or app authentication issue	~10%	See KB-007

Diagnostic Script — Session Won’t Start

Step 1 — Confirm setup and get context

- “Can I get the station ID — it’s usually a 4 to 6 digit number on a sticker on the charger?”
- “And what exactly do you see on the screen — any error codes or messages?”
- “When you plugged in, did you hear a click, or did the connector feel loose?”

Step 2 — Re-seat the cable

“First thing to try: unplug the cable completely and plug it back in firmly until you hear or feel it click. Sometimes a slightly loose connection is all it takes.”

Wait 30 seconds. If the session starts — confirm it’s running, log resolution, close call.

Step 3 — Run remote station status check

Before resetting, check the station’s real-time status in the backend:

- **Station offline / faulted:** Do not reset yet. Go to KB-003 Hardware Fault.
- **Station online / available:** Proceed to soft reset.
- **Station shows “in session”:** A ghost session may be open — clear it in the backend before resetting.

Step 4 — Soft reset

“I’m sending a remote restart to the station now — it takes about 30 seconds. Please stay plugged in and tell me when the screen changes.”

Trigger soft reset on Station ID. Wait 45 seconds. If session starts — confirm it’s running, close call.

Step 5 — Hard reset (if soft reset fails)

“The soft restart didn’t do it. I’m going to try a full power cycle on the station — this takes about 90 seconds.”

Trigger hard reset. Wait 2 minutes. If session starts — confirm, log, close.

Step 6 — If both resets fail

“The station isn’t responding to resets. Let me check something — have you tried charging at any other station recently? Did that work okay?”

- **If other stations work fine:** Fault is likely on this specific charger. Go to KB-003.
- **If other stations also fail:** Strongly suggests a vehicle-side issue. > “It’s possible the issue is actually with your car’s charging port or communication system. I’d recommend contacting your vehicle manufacturer or dealer — I know that’s not what you wanted to hear, and I’m sorry.”

Step 7 — If unresolved after 3 attempts

“I’m flagging this station for a technician inspection. I’m really sorry this didn’t get sorted today. Can I help you find the nearest available working charger while we wait?”

Offer nearest-charger lookup. Apply session credit if applicable.

Error Codes — Session Won't Start

Display Code	Meaning	Agent Action
E001 / E01	Network communication error	Soft reset; if persistent, escalate
E002 / AUTH FAIL	Authorization/ payment declined	Go to KB-002
E003 / CONNECTOR	Connector not fully seated	Ask driver to re-seat cable
E004 / VEHICLE	Vehicle not communicating	Check vehicle-side; may be car fault
E005 / GROUND	Ground fault detected	Do not reset — safety flag, escalate
OFFLINE	Charger not on network	Hard reset; if persistent, dispatch tech
RESERVED	Station reserved by another user	Check backend for open reservation
OUT OF ORDER	Hardware fault flagged	Go to KB-003

Note: Error codes vary by manufacturer (ABB, BTC Power, ChargePoint, Tritium, Wallbox, etc.). If a code isn't listed, log it verbatim for the technician.

Section B — Session Ended Early

Common Root Causes

Cause	Frequency	Identifiable by
Vehicle BMS charge limit reached	~25%	Car near full; driver set a cap
Grid fluctuation / thermal trip	~20%	Mid-session, no screen error
Payment authorization expiry	~18%	Session ends at a billing interval
Handshake dropout (ISO)	~15%	Screen shows “communication error”

Cause	Frequency	Identifiable by
15118 failure)		
Network/connectivity loss at charger	~10%	Screen goes dark or shows “offline”
Hardware thermal protection trip	~8%	Hot weather, extended session
Vehicle-side fault	~4%	Vehicle dashboard warning light

Diagnostic Script — Session Ended Early

Step 1 — Get timeline and symptoms

“How far into the session was it when it stopped — roughly how many minutes or what percentage?”

“Did anything show on the screen when it stopped, or did it just go quiet?”

“Did you see any warning lights on your car’s dashboard?”

Decision tree based on timing:

Under 2 minutes: Likely a payment authorization failure or handshake failure. > “An early stop like that is usually a payment authorization issue. Let me check the session on our end.” - If payment failed: Go to KB-002. - If handshake error: Soft reset and retry.

2–20 minutes in, no screen message: Likely a grid fluctuation or network dropout. > “A drop in the middle like that can be a brief power grid fluctuation or a temporary connectivity hiccup at the station. I’m going to send a reset and you can try starting a new session — if it drops again at the same point, we’ll get a tech out.”

Mid-session with a vehicle dashboard warning: Vehicle-side issue. > “That dashboard warning is your car telling you something — it may have stopped the session as a protective measure. I’d suggest checking your vehicle manual or contacting your dealer. The charger may be fine on our end.”

Session showed “complete” but battery isn’t full: Charge limit set in vehicle. > “It’s worth checking in your vehicle’s app or settings — a lot of drivers set a charge cap at 80 or 90 percent to protect their battery. If there’s a cap set, that would explain this.”

Step 2 — Retry session

“Let me send a reset to the station and you can try starting a new session. Please let me know if it drops again.”

Step 3 — If session drops a second time in the same phase

“Dropping twice in the same spot suggests a hardware or firmware issue at this station. I’m going to flag it for a technician. I’m really sorry about the inconvenience.”

Step 4 — Billing review for incomplete session

“If you were charged for a session that didn’t complete, I’ll flag that for our billing team. You shouldn’t pay for a session that didn’t deliver. Would you like me to note that now?”

Log: session start time, end time, kWh delivered (if available), charge amount, reason for early stop.

Section C — Repeated Failures (Same Driver, Multiple Locations)

If a driver reports that sessions consistently fail to start or end early across multiple chargers on different networks:

1. The fault is almost certainly vehicle-side — specifically the CCS inlet, the vehicle BMS, or the onboard charger.
2. Do not dispatch a technician to the charging station.
3. Script: > “Given that you’re seeing this at more than one station, the issue is likely with your vehicle’s charging system rather than the chargers themselves. I’d strongly recommend a visit to your dealership or an EV-certified mechanic. They can run a diagnostic on your charging port and battery management system.”
4. Log the vehicle make, model, and year for pattern tracking.

Escalation Criteria

Condition	Action
Hard reset fails, station shows online	Escalate to Tier 2
Same station fails for 3+ drivers in same day	Priority technician dispatch
Vehicle dashboard shows electrical fault	Direct to vehicle manufacturer
Driver was charged for a session that never started	Billing flag + confirm refund timeline
Driver is stranded, no nearby alternatives	Tier 2 + session credit

Agent Reminders

- Always check station status before resetting — resetting an already-faulted unit can corrupt logs.
- If a session has a ghost “in progress” state in the backend, clear it before attempting a new session.
- Never tell a driver their car is broken — say “it may be worth checking the vehicle’s charging system.”
- If the driver was charged for a failed session, flag billing before they hang up — do not make them call back.